



Validation Strategies

SOUTHWEST RESEARCH INSTITUTE®

Andrés Muñoz-Jaramillo

andres.munoz@swri.org

Commissioned and funded by NASA's Office of the Chief Science Data Officer (OCSDO)



SOLAR SYSTEM SCIENCE & EXPLORATION

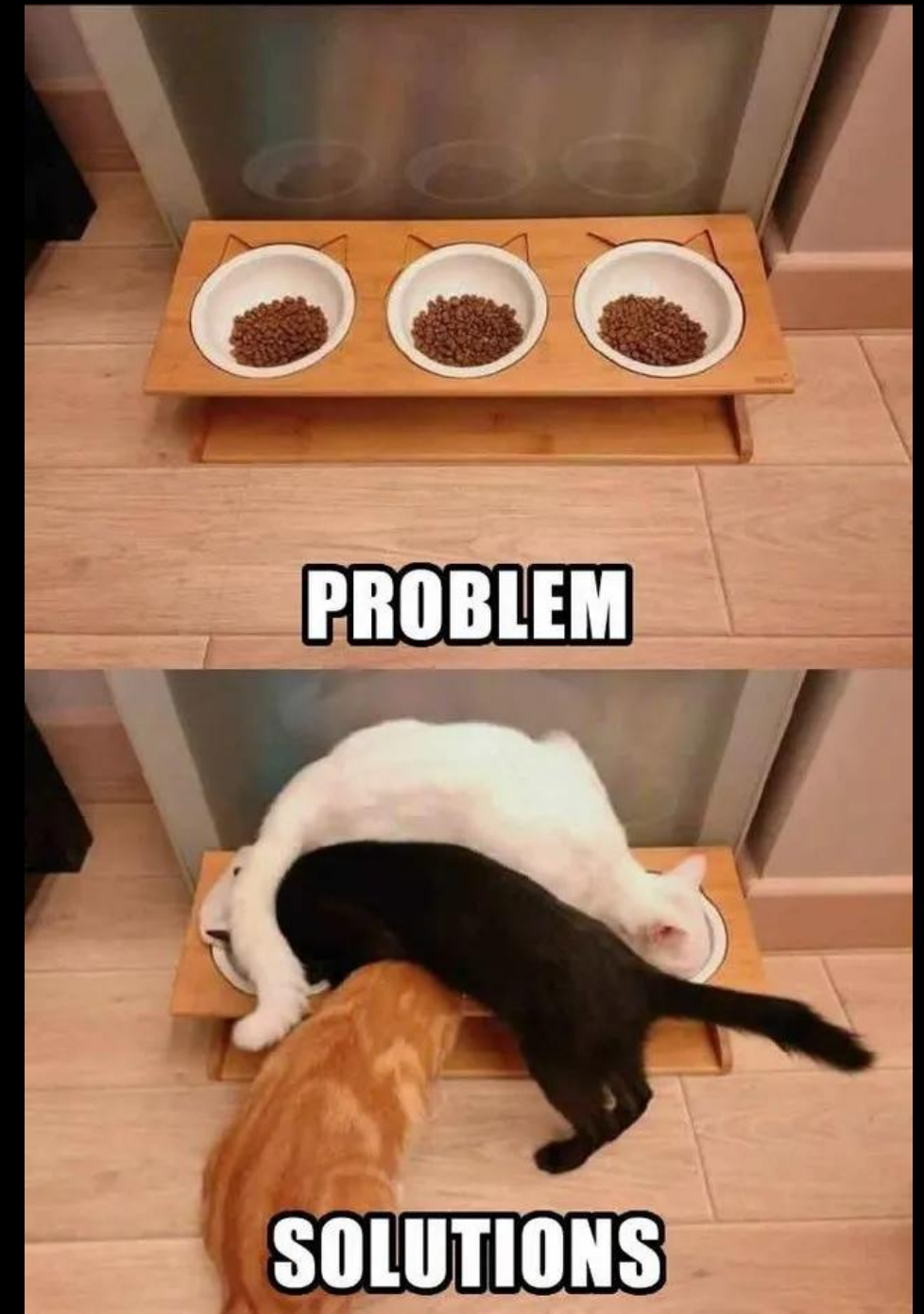
Some things to keep in mind

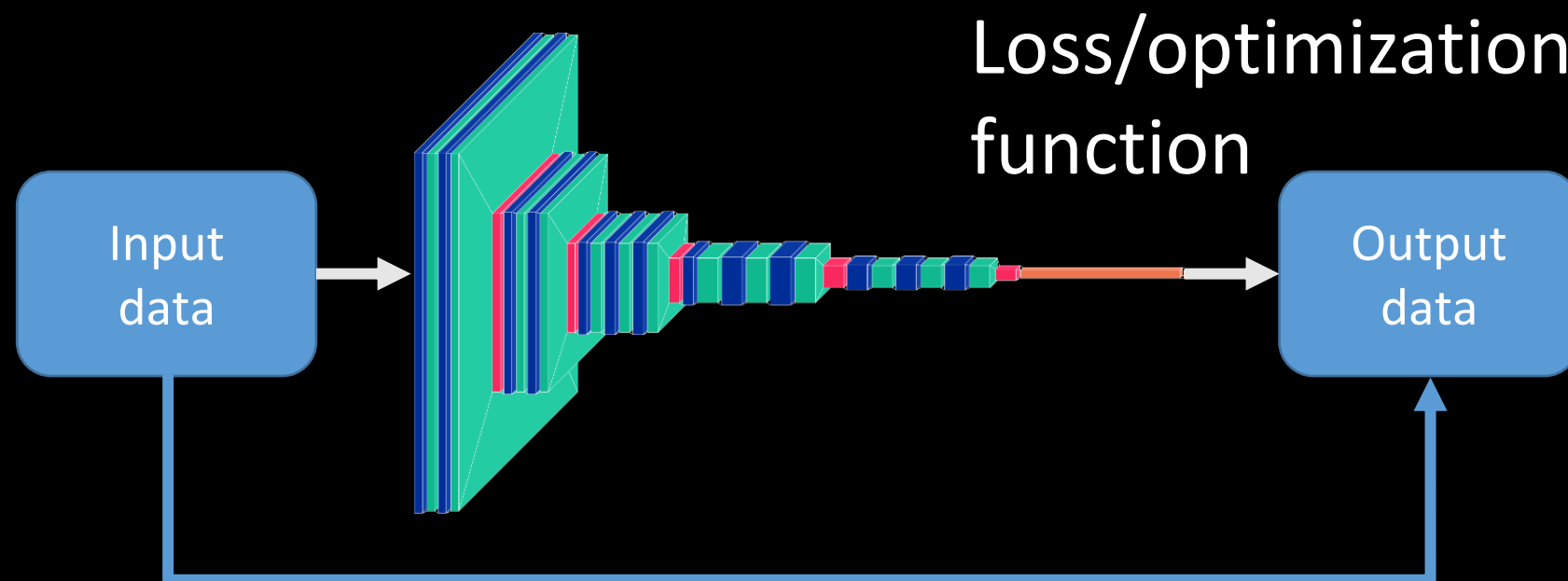
- If you are a virtual participant, use your breakout rooms and voice.
- I pushed a training script for both baselines and finetune.
- Let's plan for a stress test before the panel.



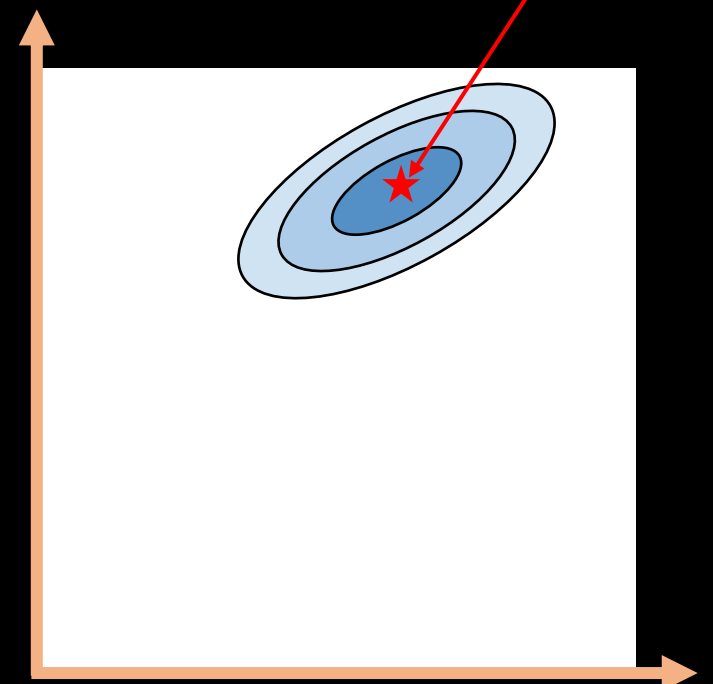
- Training deep learning models involves solving a problem we can solve (**the past**) in the hopes that the solution applies to the problem we want to solve, but can't (**the future**).
- There is a strong incentive (and it's very easy) for the human-in-the-loop to muddle and/or increase disparity between the solving for the past and solving for the future.
- Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:
 - Using ensembles.
 - Data splitting and data management.
 - Using the right metrics.
 - Using simple baselines.

Training deep learning models involves solving a problem we can solve (the past) in the hopes that the solution applies to the problem we want to solve, but can't (the future)



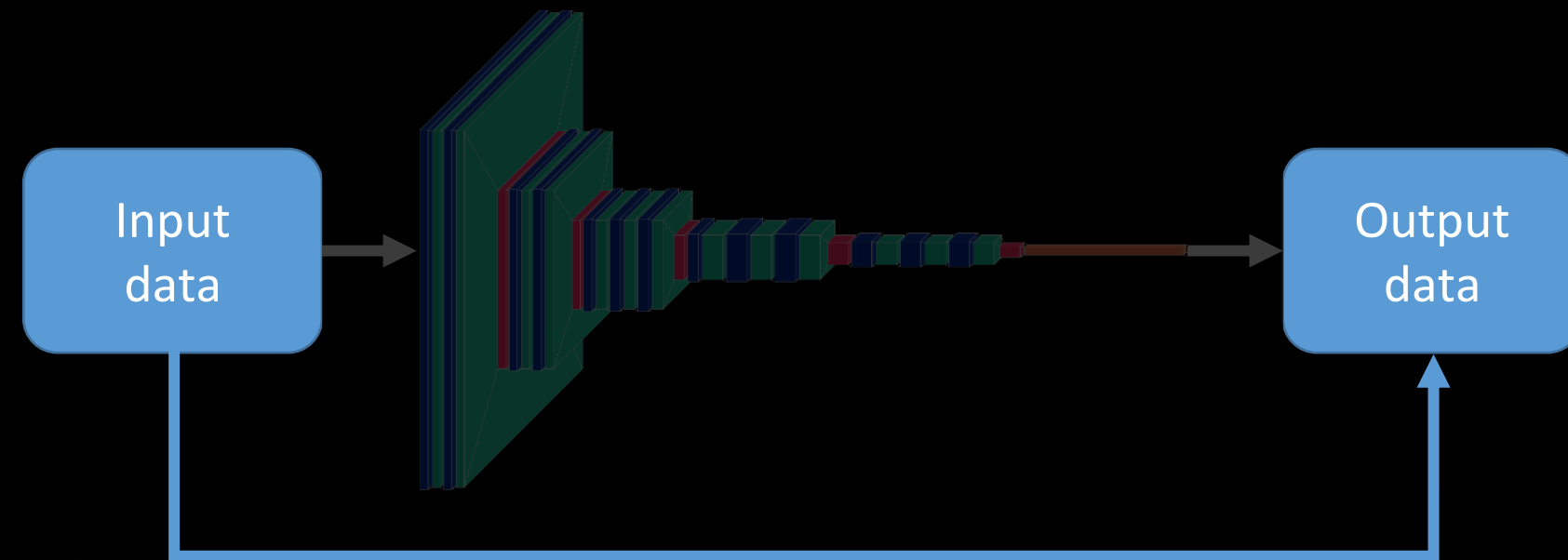


This is the real goal!

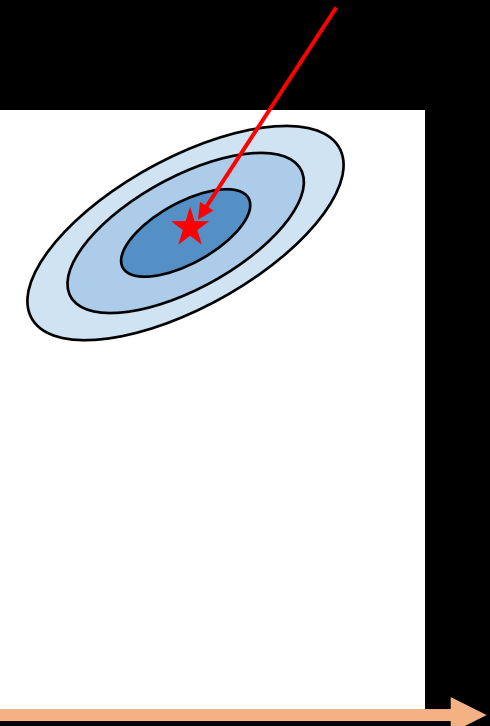


This is not a problem we can solve

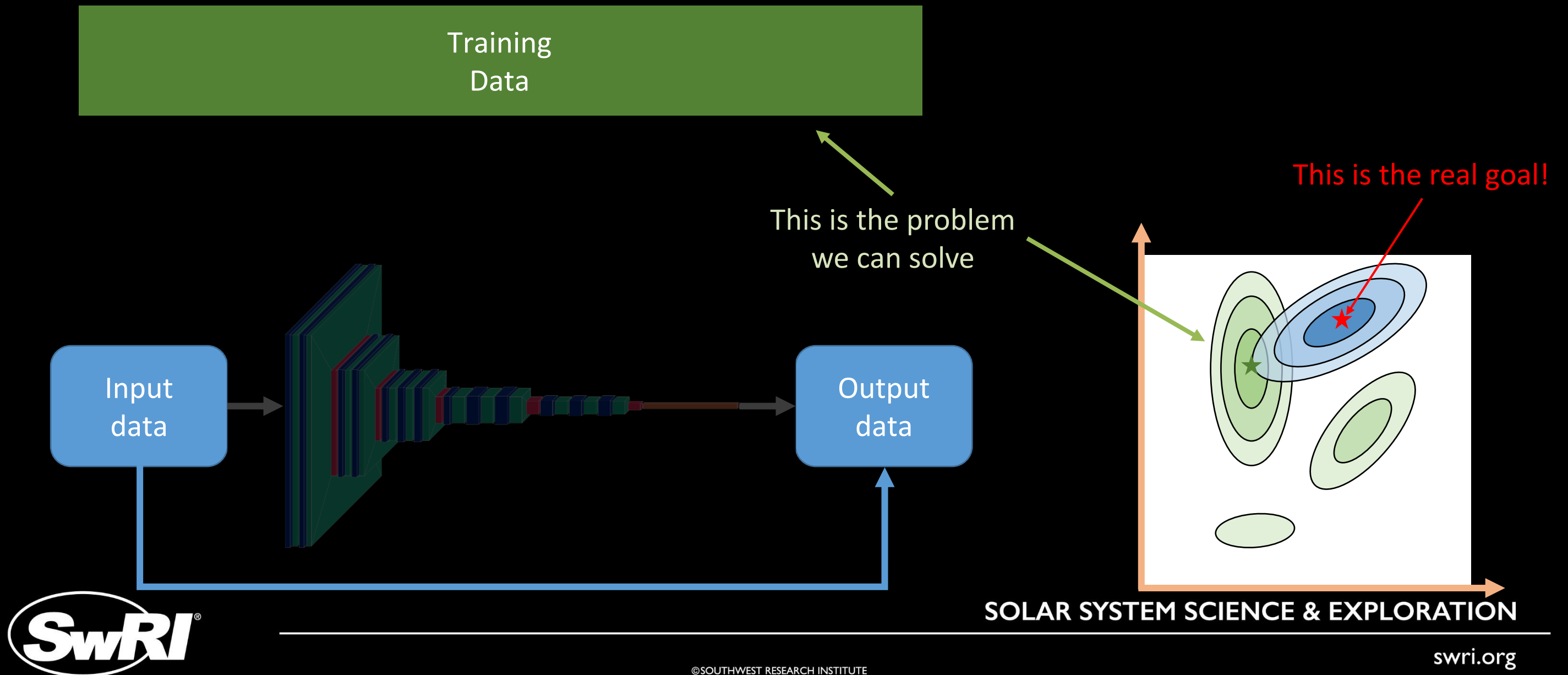
Training
Data



This is the real goal!



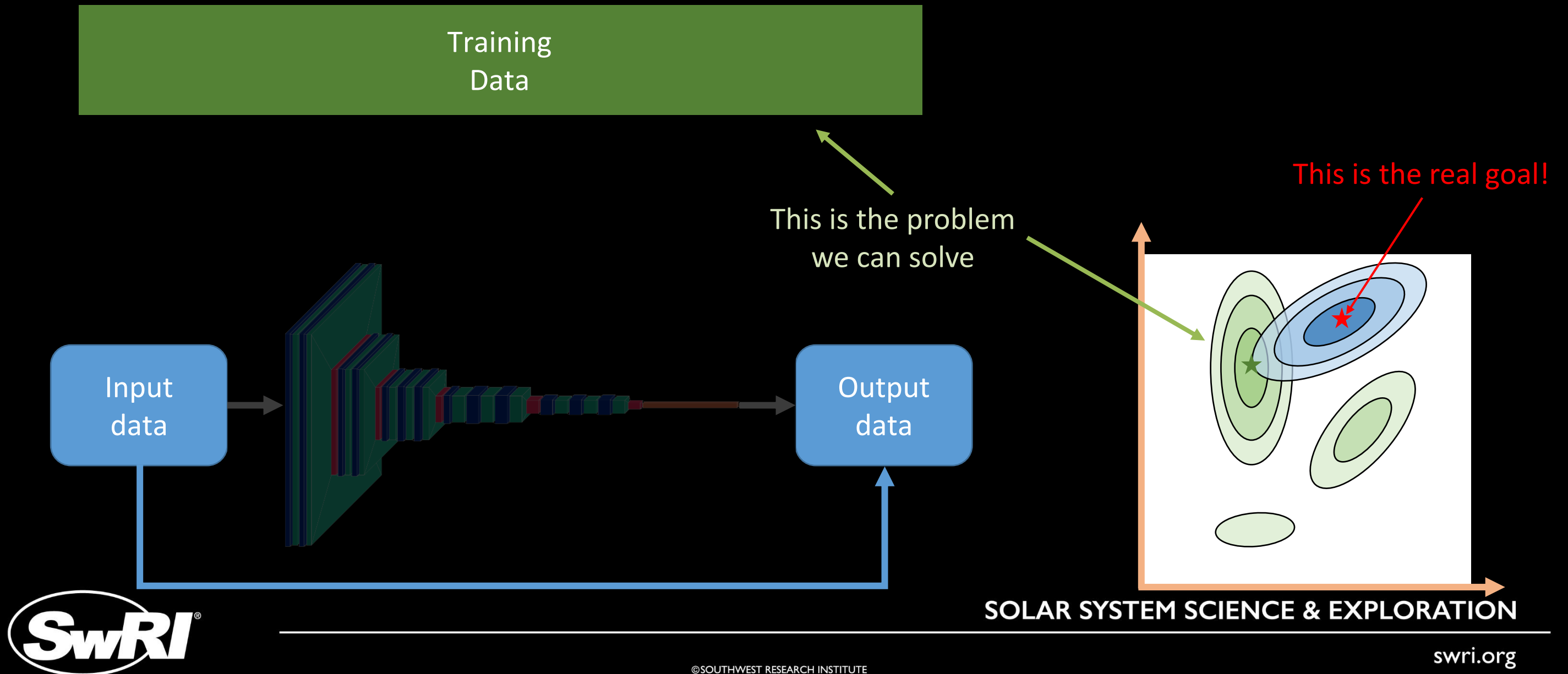
This is not a problem we can solve



There is a strong incentive (and it's very easy) for the human-in-the-loop to muddle and/or increase disparity between the solving for the past and solving for the future.

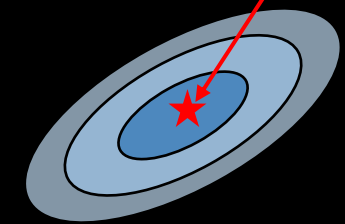


If we choose the wrong training data, our results will not be useful



If we choose the wrong training data, our results will not be useful

This is the real goal!

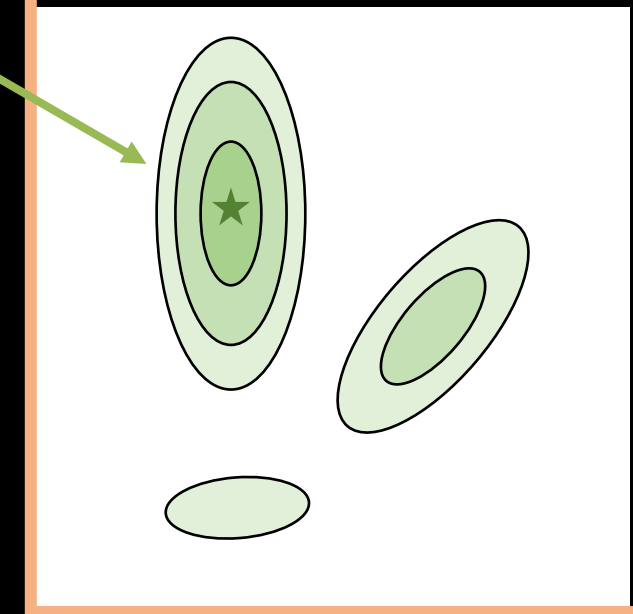


Training
Data

This is the problem
we can solve

Input
data

Output
data

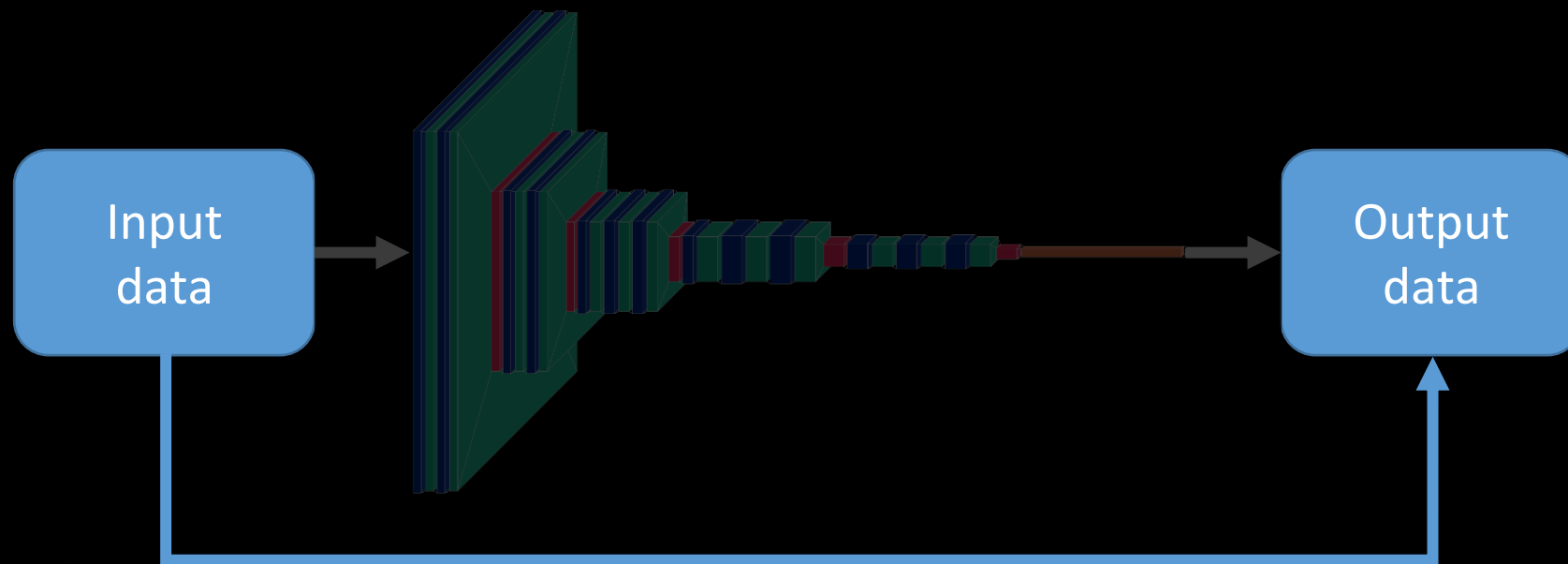


SOLAR SYSTEM SCIENCE & EXPLORATION

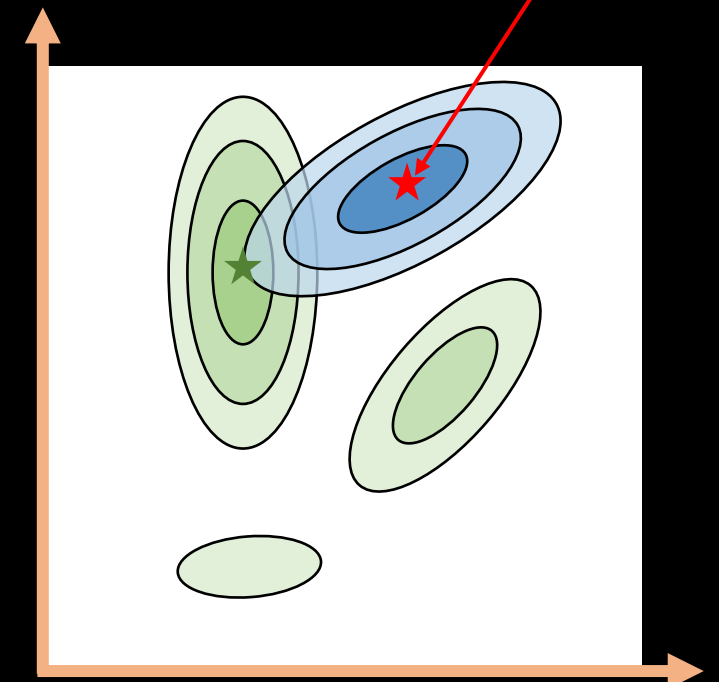


If we choose the wrong test set, our results will not be useful

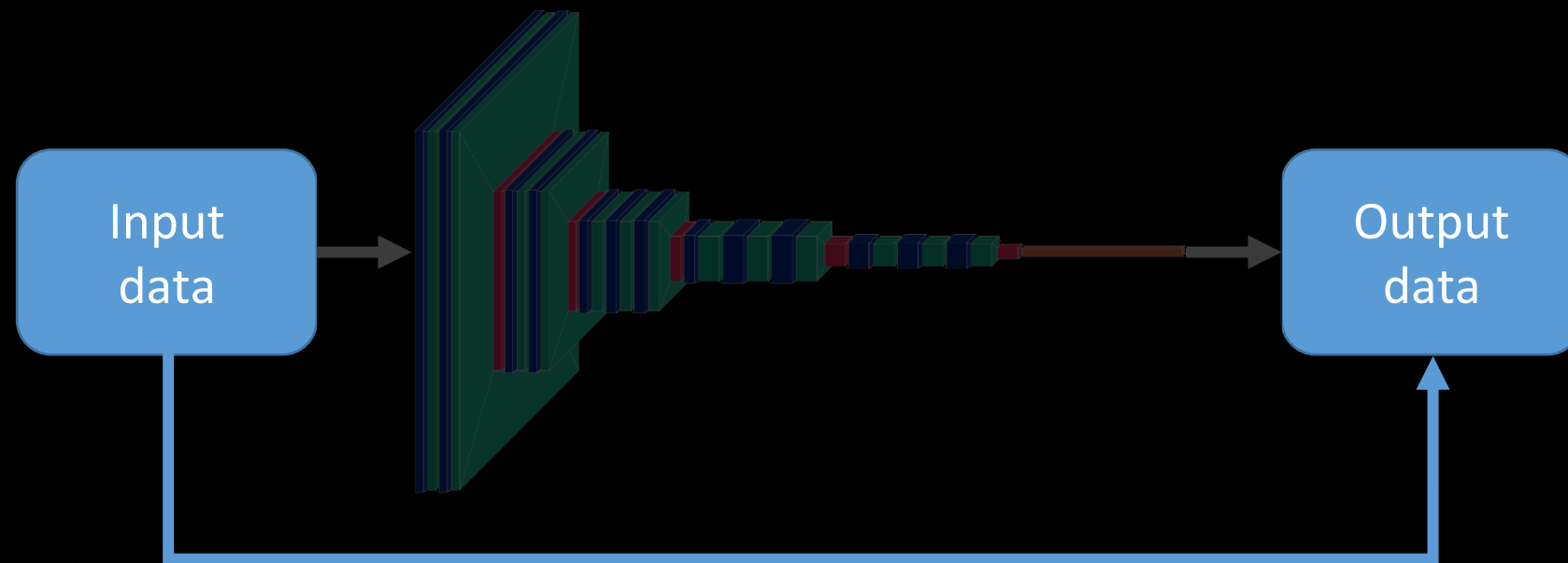
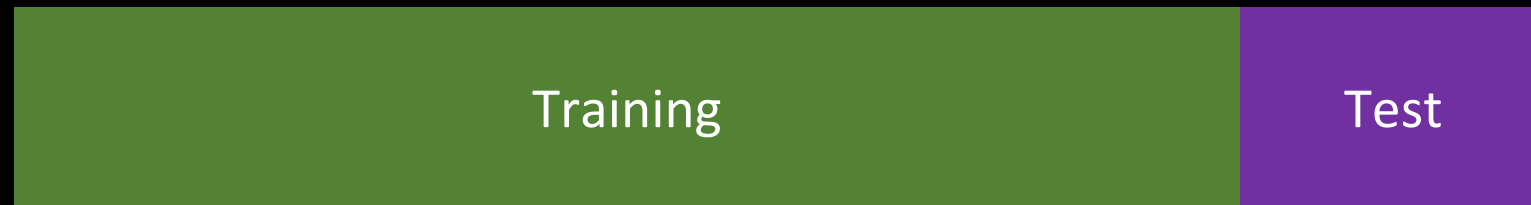
Training
Data



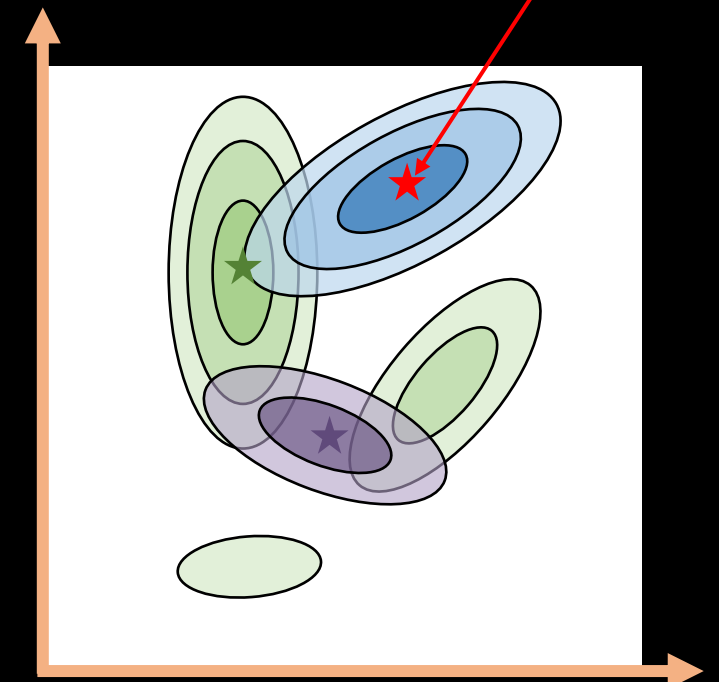
This is the real goal!



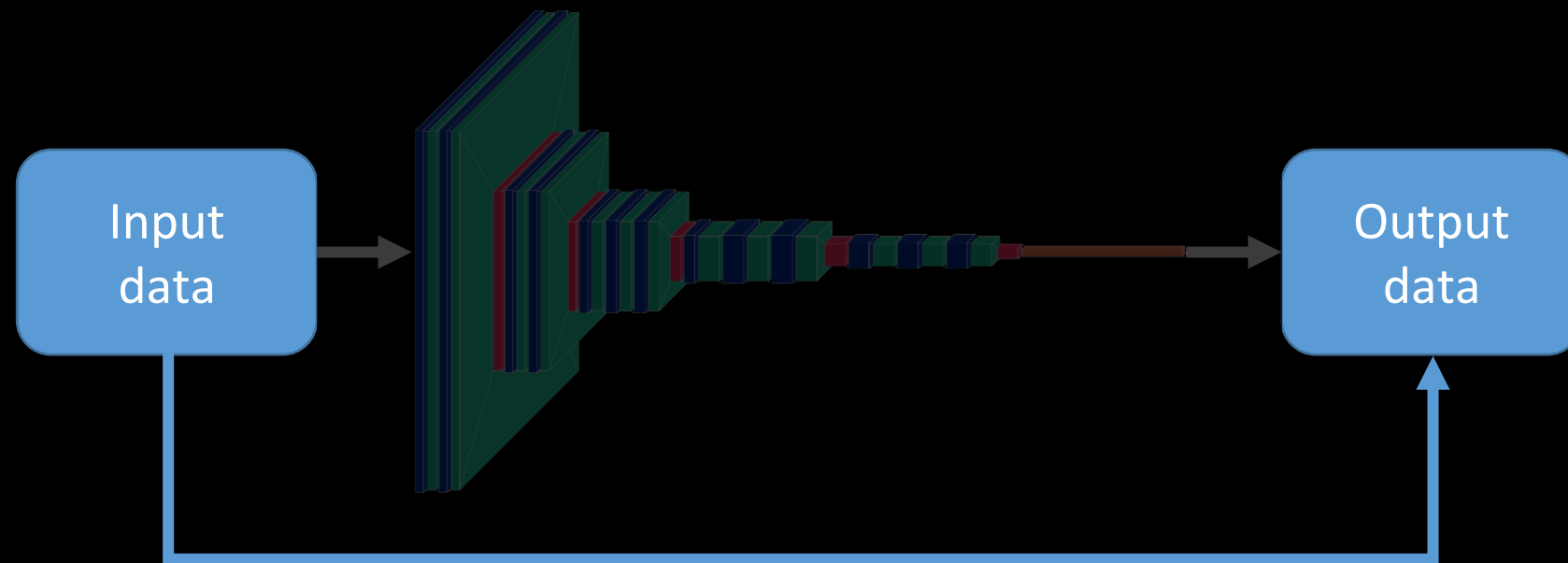
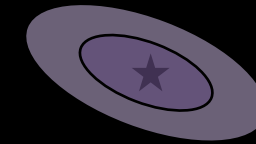
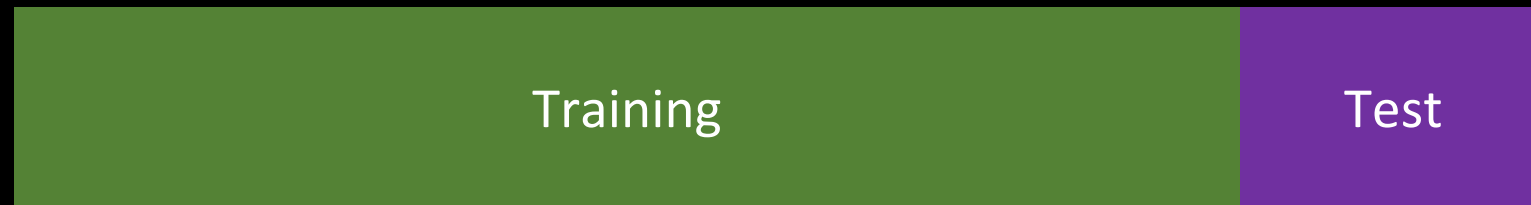
If we choose the wrong test set, our results will not be useful



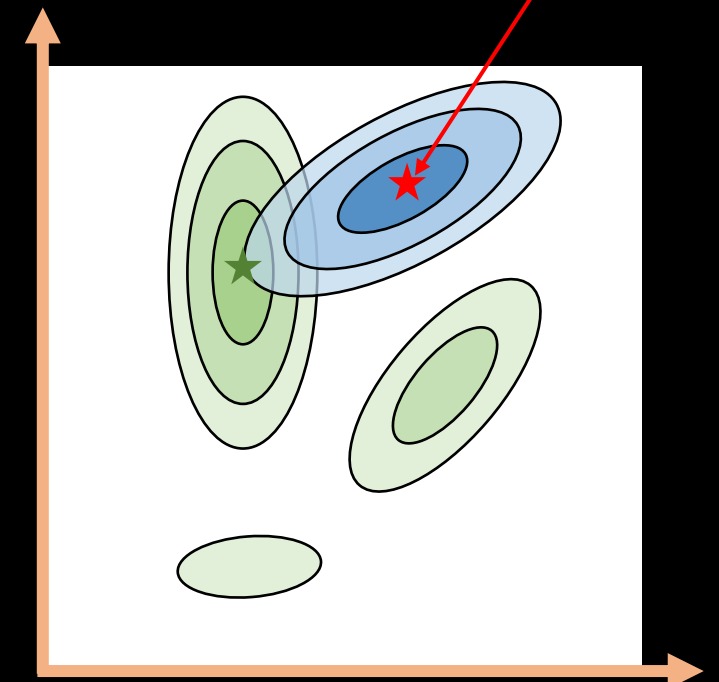
This is the real goal!



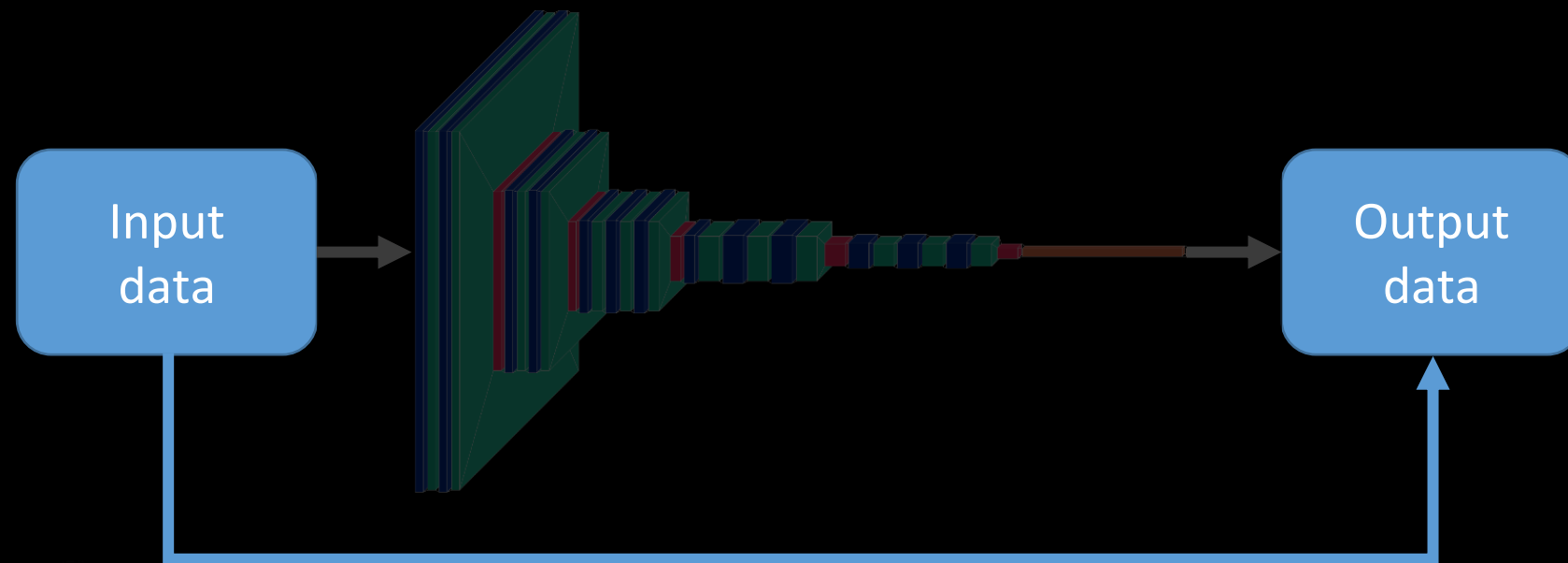
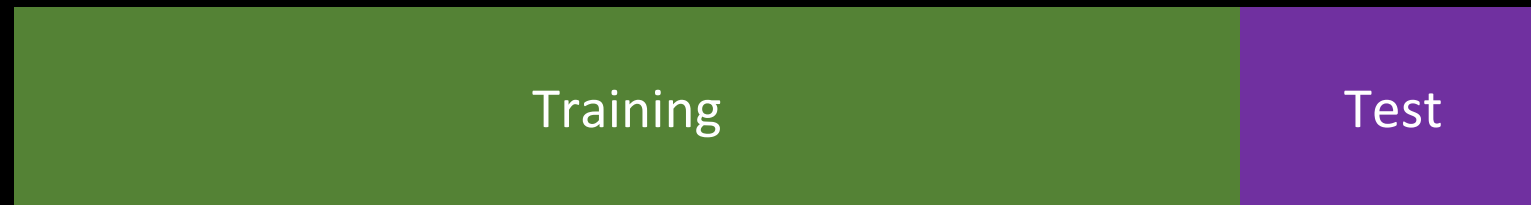
If we choose the wrong test set, our results will not be useful



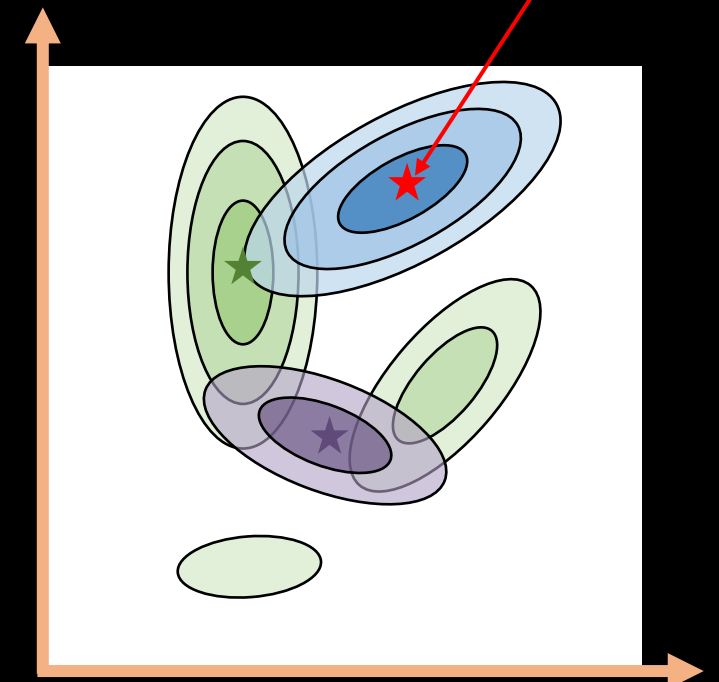
This is the real goal!



If we overfit our test through our design choices, our result will not be useful

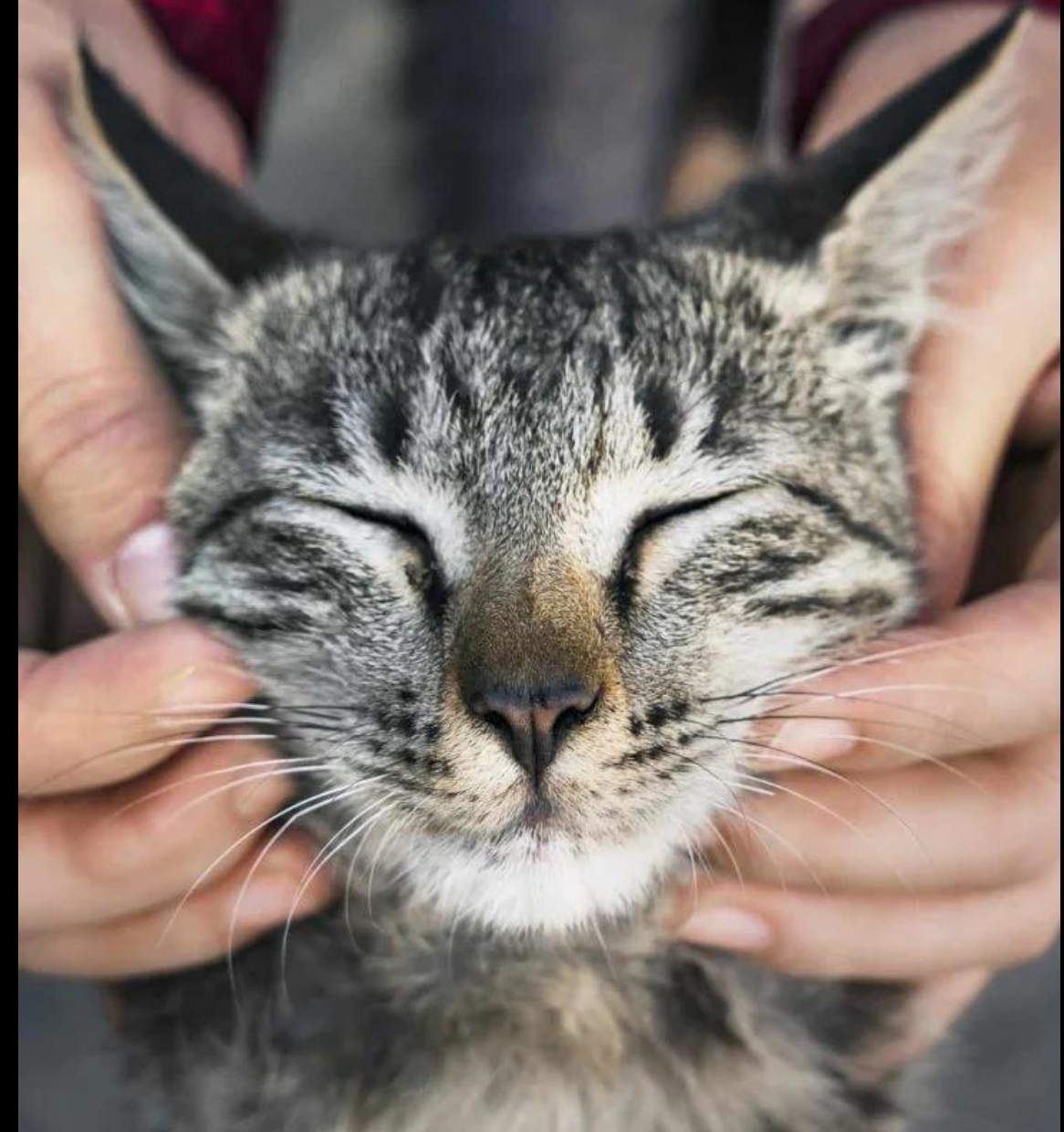


This is the real goal!



Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:

- Using ensembles.**
- Data splitting and data management.**
- Using the right metrics.**
- Using simple baselines.**



Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:

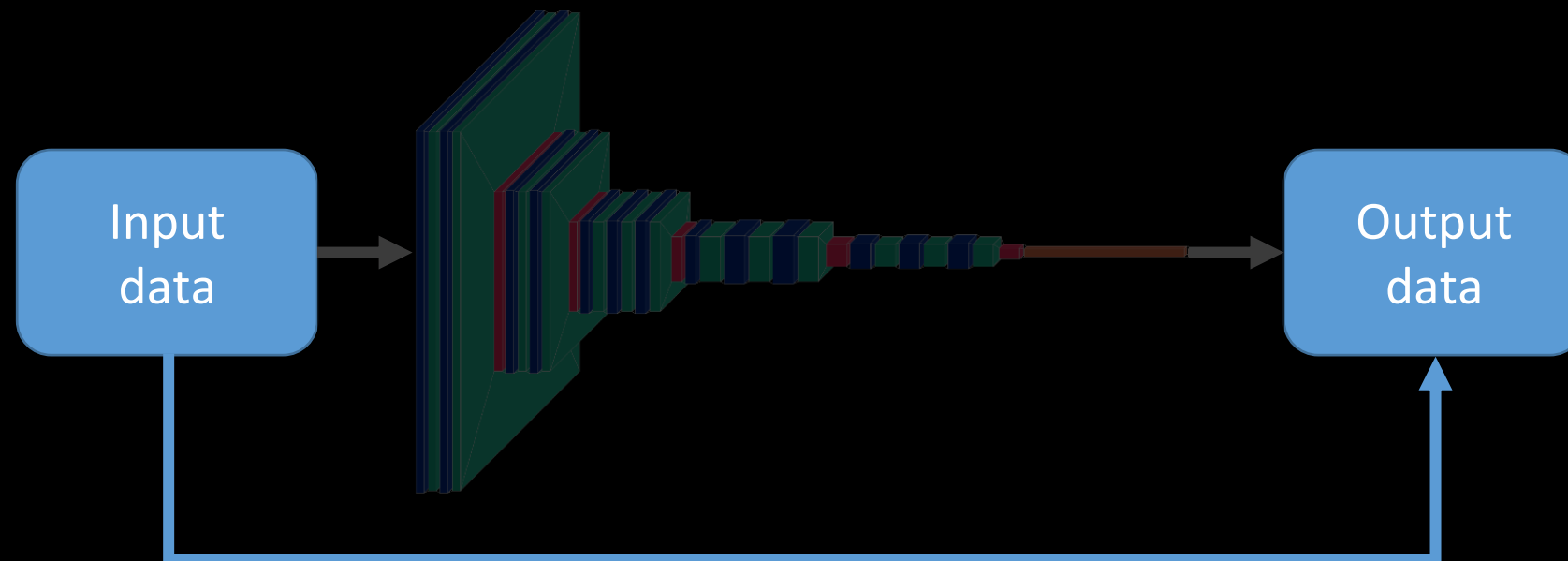
- Using ensembles.**
- Data splitting and data management.**
- Using the right metrics.**
- Using simple baselines.**



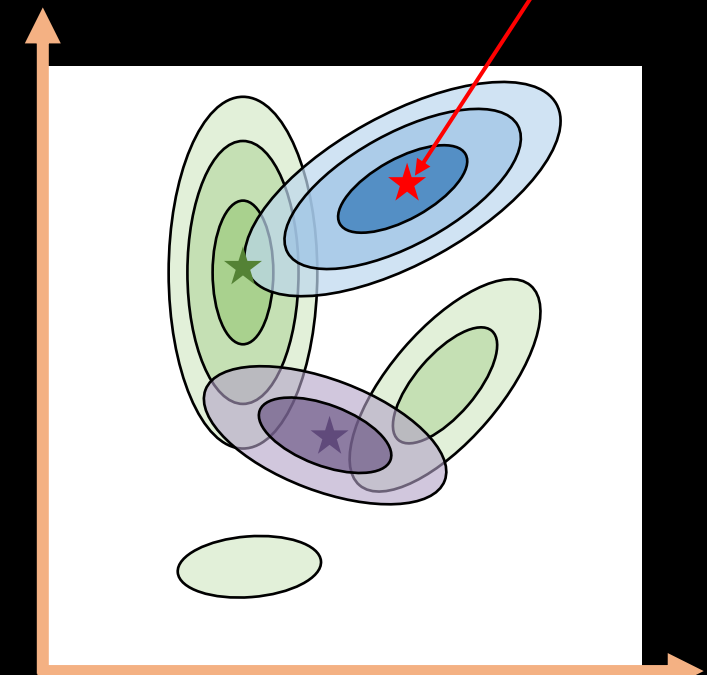
Are we finding the right minimum?

Training

Test



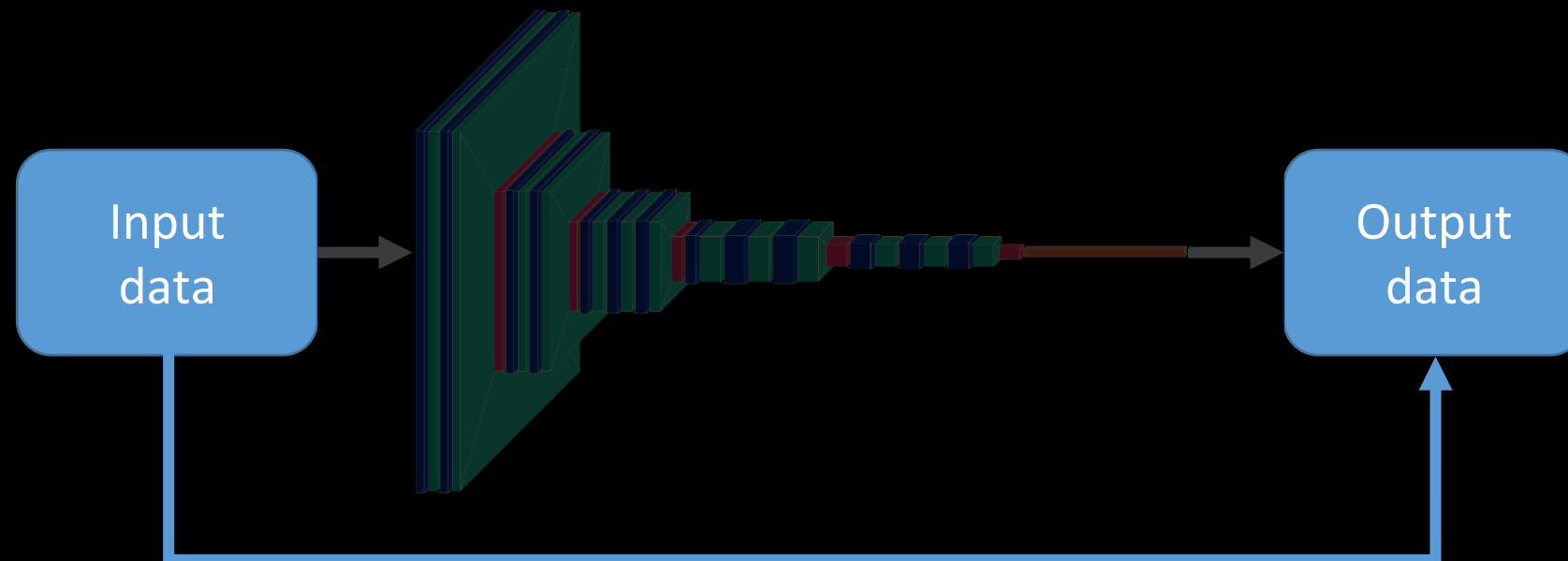
This is the real goal!



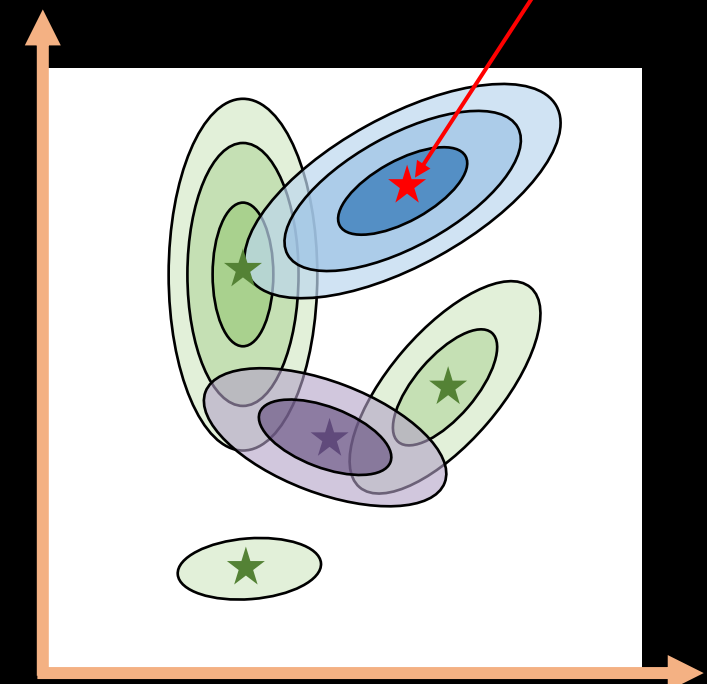
Are we finding the right minimum?

Training

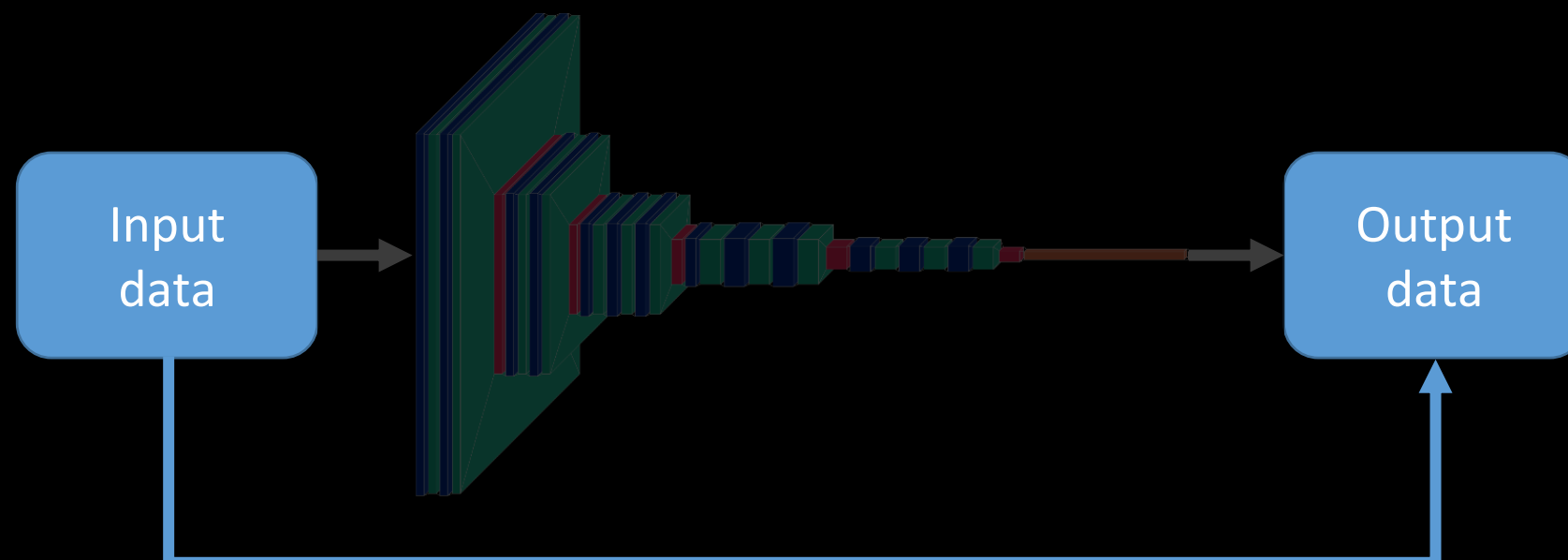
Test



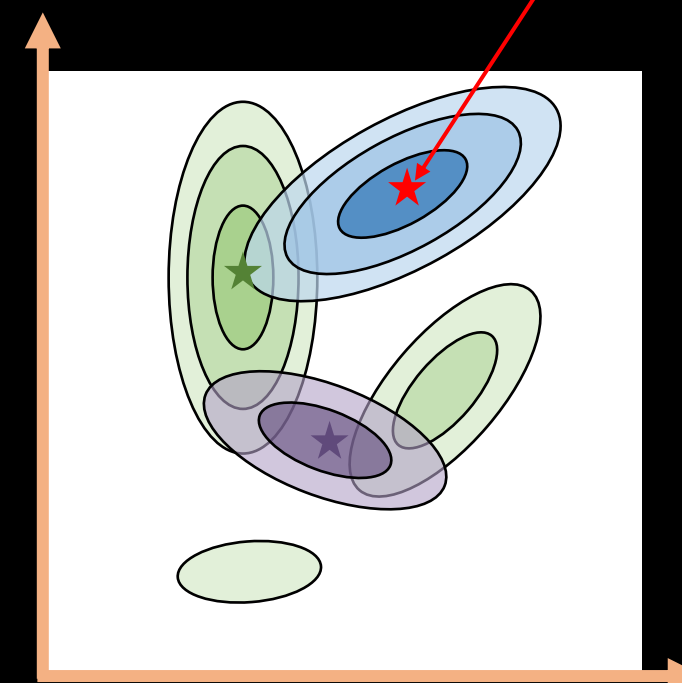
This is the real goal!



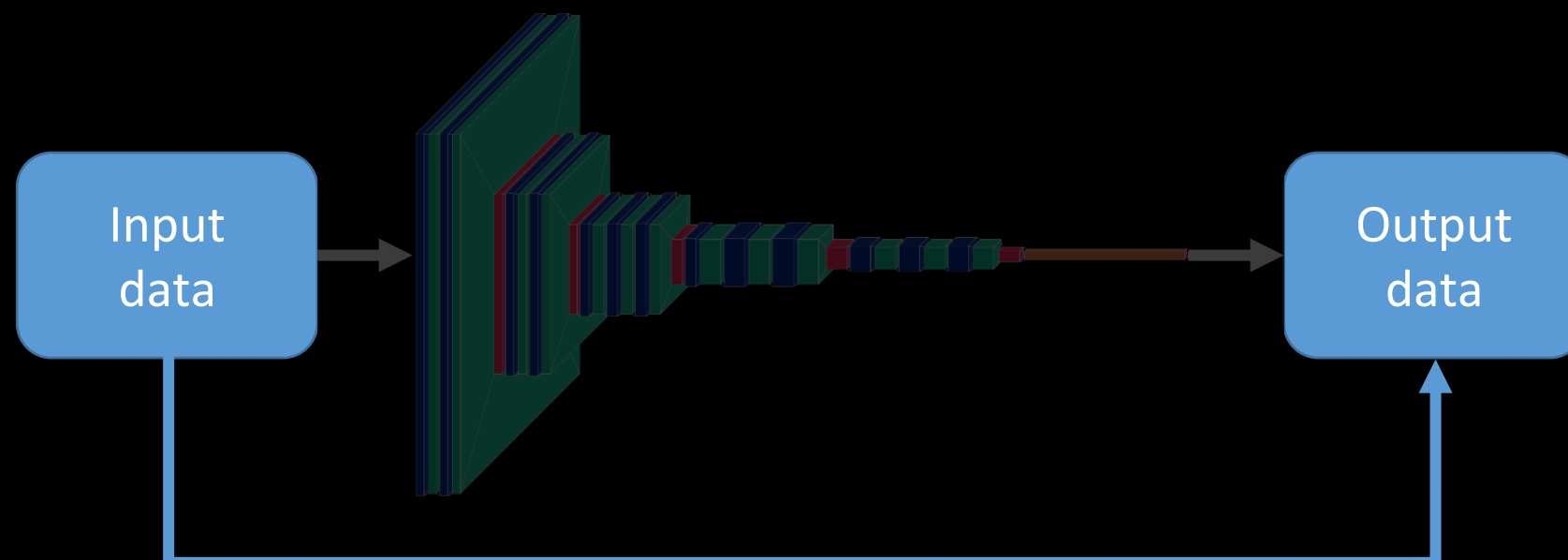
Are we sensitive to data splitting?



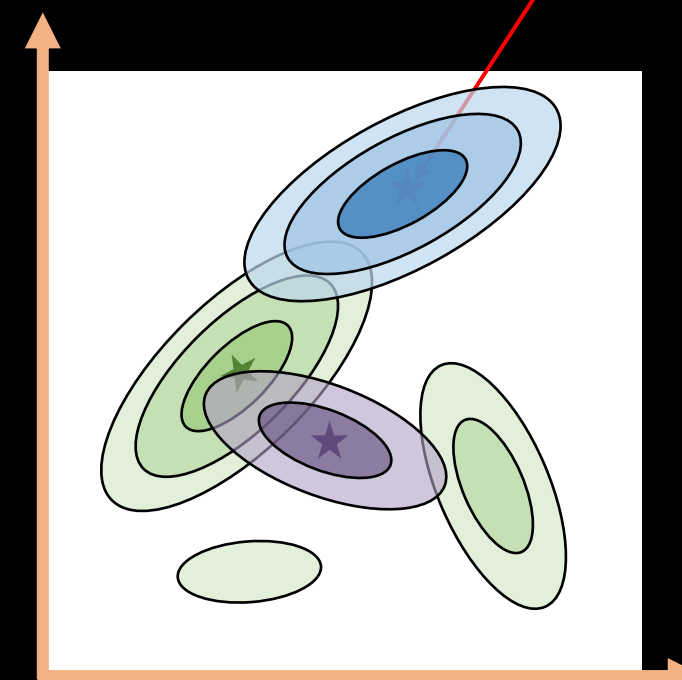
This is the real goal!



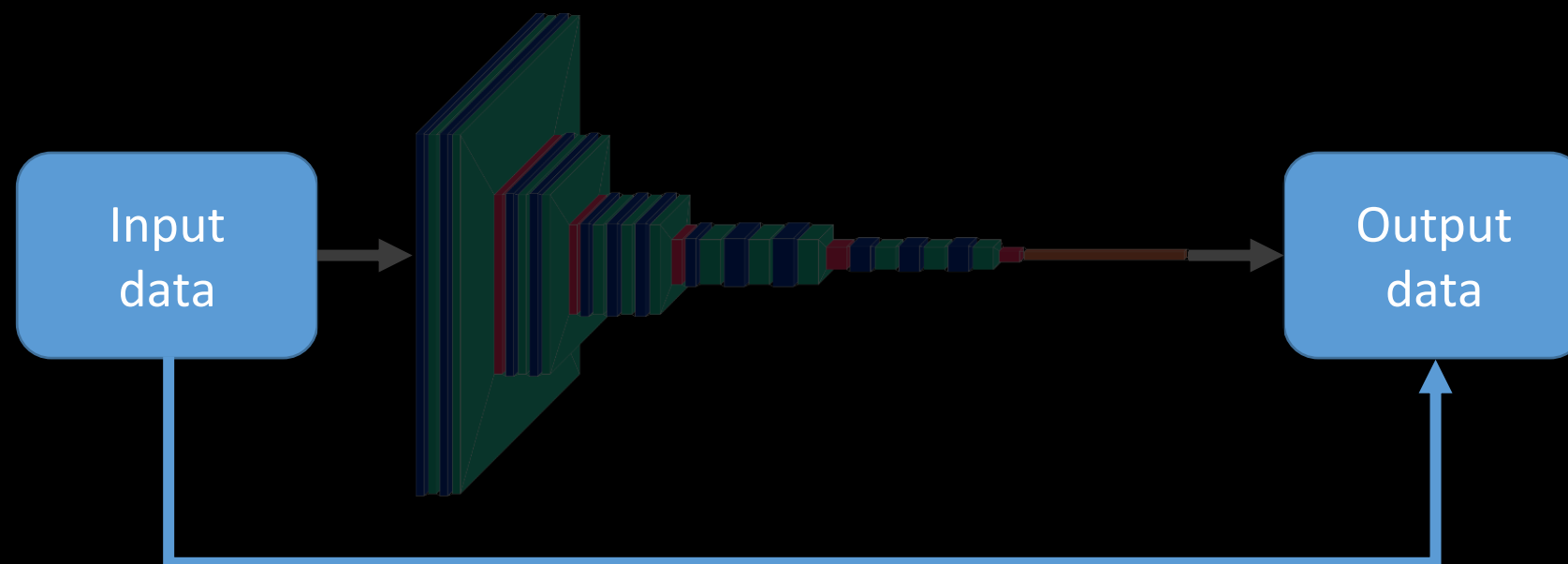
Are we sensitive to data splitting?



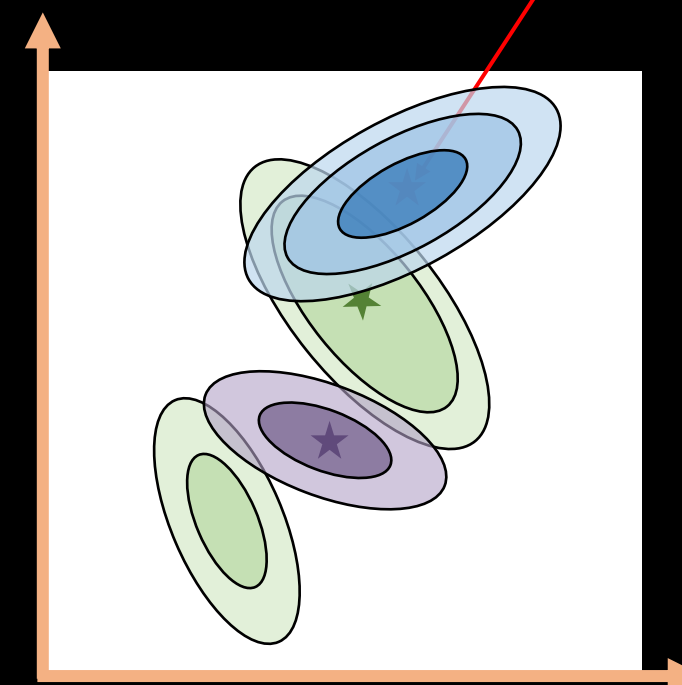
This is the real goal!



Are we sensitive to data splitting?



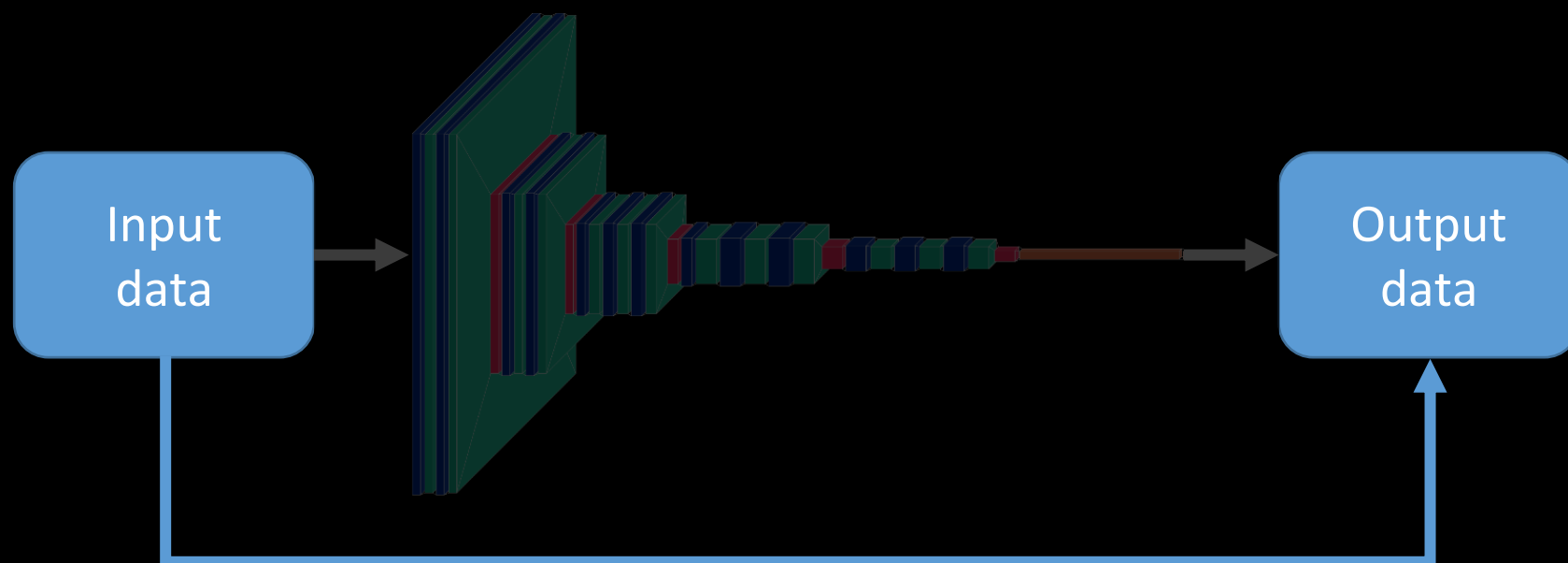
This is the real goal!



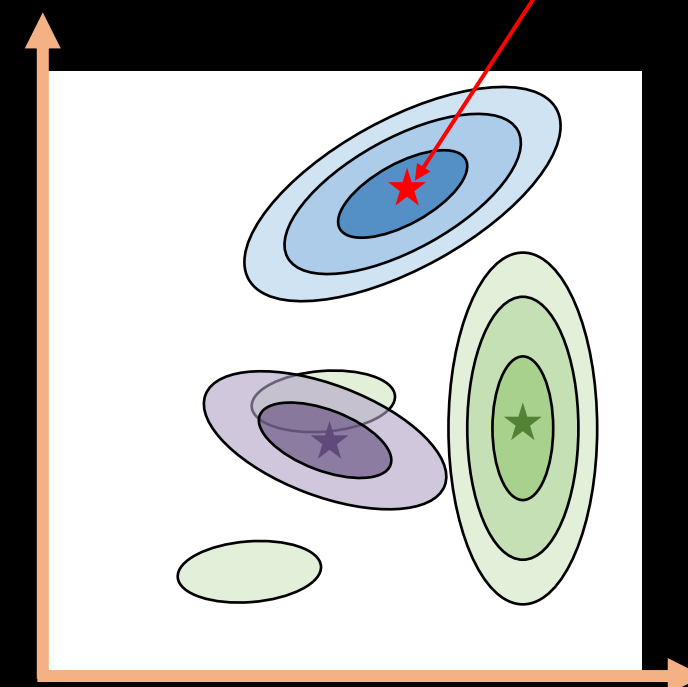
Are we sensitive to data splitting?

Validation

Test



This is the real goal!



What is the answer to the Ultimate Question of Life, the Universe, and Everything?



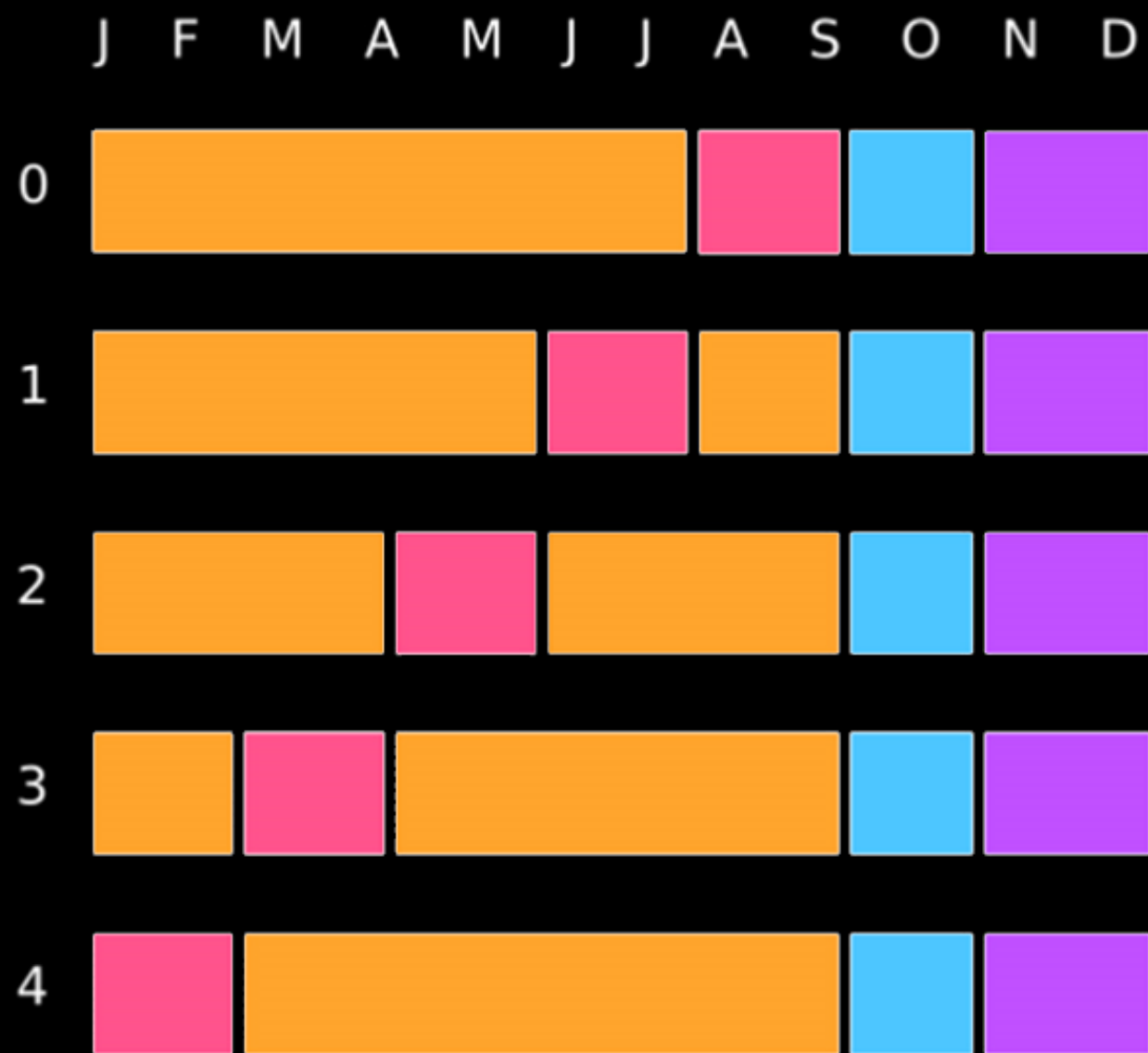
We are better off harnessing the wisdom of the crowd! And it helps with human overfitting!



Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:

- Using ensembles.
- **Data splitting and data management.**
- Using the right metrics.
- Using simple baselines.





Yearly Data Split

Training



Validation



Hold-out
4.48

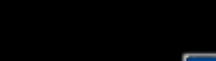


Test A
3.76



Cycle
Agnostic
Test

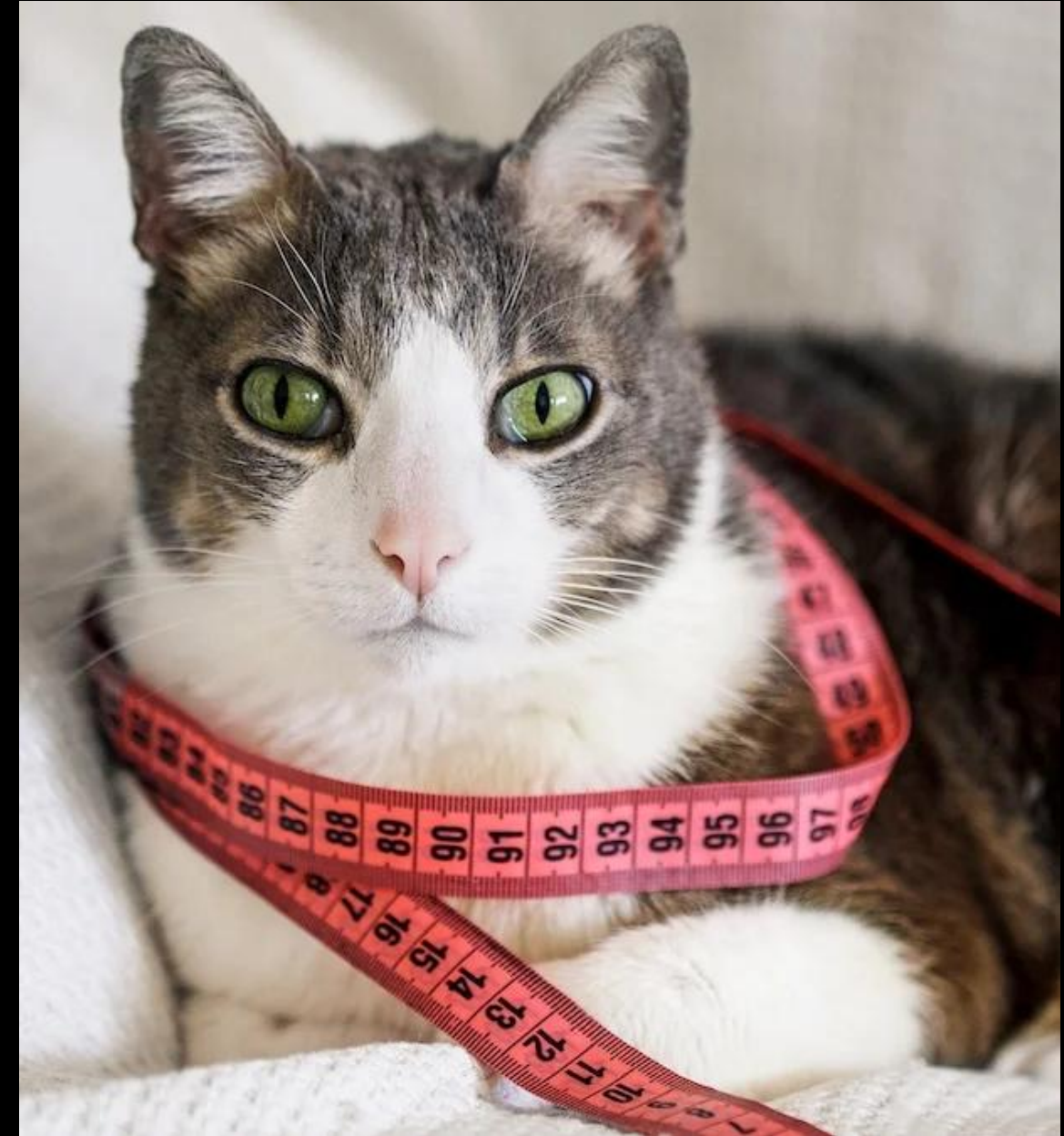
Test B
27.24



Operational
Test
(2016–2017)

Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:

- Using ensembles.
- Data splitting and data management.
- **Using the right metrics.**
- Using simple baselines.



Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:

- Using ensembles.
- Data splitting and data management.
- Using the right metrics.
- **Using simple baselines.**



- Deep learning is a form of machine learning that has shown great success at approximating complex relationships between data.
- Training deep learning models involves solving a problem we can solve (the past) in the hopes that the solution applies to the problem we want to solve, but can't (the future).
- There is a strong incentive (and it's very easy) for the human-in-the-loop to muddle and/or increase disparity between the solving for the past and solving for the future.
- Best practice protocols (and iron discipline) are critical to ensure that we produce useful models and harness the potential of deep learning:
 - Using ensembles.
 - Data splitting and data management.
 - Using the right metrics.
 - Using simple baselines.

Protocol for Deep learning applications to Space Weather

- Built on/consolidating suggestions in Barnes et al. (2016); Jonas et al. (2018); Leka et al. (2019a,b); Park et al. (2020); Angryk et al. (2020); Ahmadzadeh et al. (2021); Guastavino et al. (2022); Deshmukh et al. (2023); & van der Sande et al. (2023a).
- <https://doi.org/10.5281/zenodo.8408373>
- <https://github.com/SwRI-IDEA-Lab/sw-forecast-protocol>
- Data split, ensemble learning, testing discipline, baselining, metrics, and probability calibration.



Choose your own adventure

- Runs with different amounts of training data and the same validation.
- Runs with different training sets and the same validation (build your own ensemble).
- Baseline and finetuning runs with identical training and validation sets.